



CSA0810-PYTHON PROGRAMMING

Automated Social Media Sentiment Normalization Tool with CLI, Textual Histogram & SQL Integration

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ABSTRACT



- Automates the analysis of sentiment in social media text.
- Normalizes text to improve data quality and consistency.
- Classifies content into positive, negative, and neutral sentiments.
- Provides **command-line execution** for simple and efficient operation.
- Generates a **textual histogram** to show sentiment distribution.
- Uses **SQL integration** for storing and managing analyzed data.
- Provides a fast, reliable, and user-friendly sentiment analysis solution.



INTRODUCTION



- An automated tool designed to analyze and normalize sentiment from social media text.
- It processes textual data and identifies positive, negative, and neutral sentiments.
- The tool uses command-line execution for simple and efficient user interaction.
- Text normalization improves the quality and consistency of social media data.
- A textual histogram visually represents sentiment distribution in a simple format.
- SQL integration enables efficient storage, retrieval, and management of analyzed data.
- The system provides a fast, reliable, and user-friendly solution for sentiment analysis.



PROBLEM STATEMENT

- Social media generates a large amount of unstructured textual data every day.
- Raw social media text often contains slang, symbols, spelling variations, and unnecessary words.
- Manual sentiment analysis is time-consuming and difficult for large datasets.
- There is a need to automatically classify text into positive, negative, and neutral sentiments.
- Traditional analysis methods may not provide a simple visual representation of sentiment distribution.
- Managing and retrieving analyzed sentiment data efficiently requires database integration.
- Therefore, an automated tool is needed to **normalize, analyze, visualize, and store social media sentiment data efficiently.**

LITERATURE REVIEW

S.No	Year	Area / Study	Key Findings	Limitation / Gap
1	2024	Sentiment Analysis	Identifies positive, negative, and neutral opinions.	Depends on text quality.
2	2025	Text Normalization	Removes noise, slang, symbols, and unnecessary words.	Social media language is challenging.
3	2022	Machine Learning	Supports automatic sentiment classification.	Requires suitable training data.
4	2023	Social Media Analysis	Helps understand public opinions and trends.	Large datasets are difficult to process.
5	2024	SQL & Visualization	Stores results and displays sentiment distribution effectively.	Requires proper database management.



OBJECTIVES



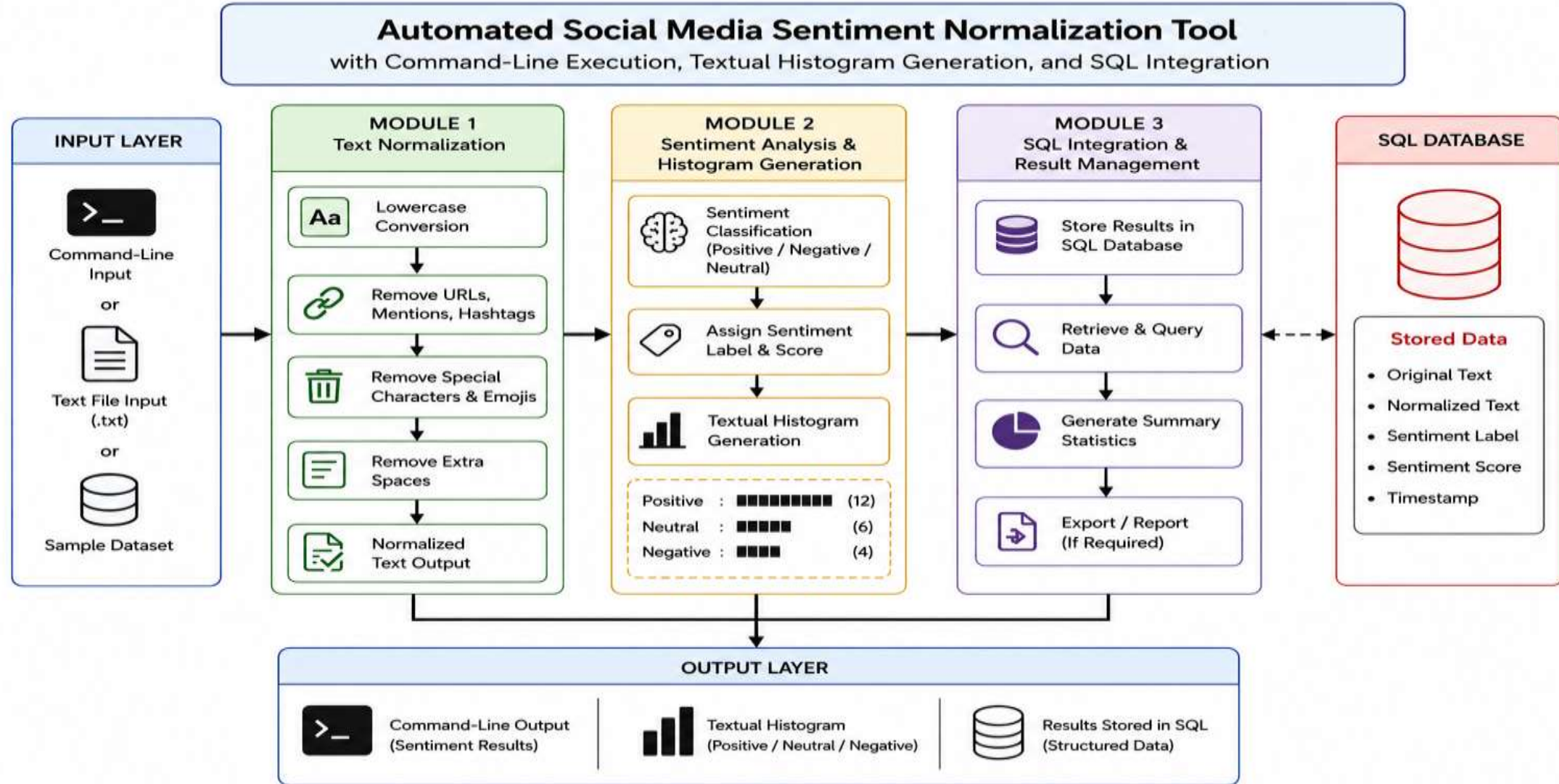
- To develop an automated system for **social media sentiment analysis and text normalization**.
- To clean and preprocess text by removing **noise, symbols, and unnecessary words**.
- To classify social media content into **positive, negative, and neutral sentiments**.
- To provide a simple **command-line interface (CLI)** for efficient execution.
- To generate **textual histograms** for easy visualization of sentiment distribution.
- To use **SQL integration** for storing and retrieving sentiment analysis results efficiently.



Engineer to Excel

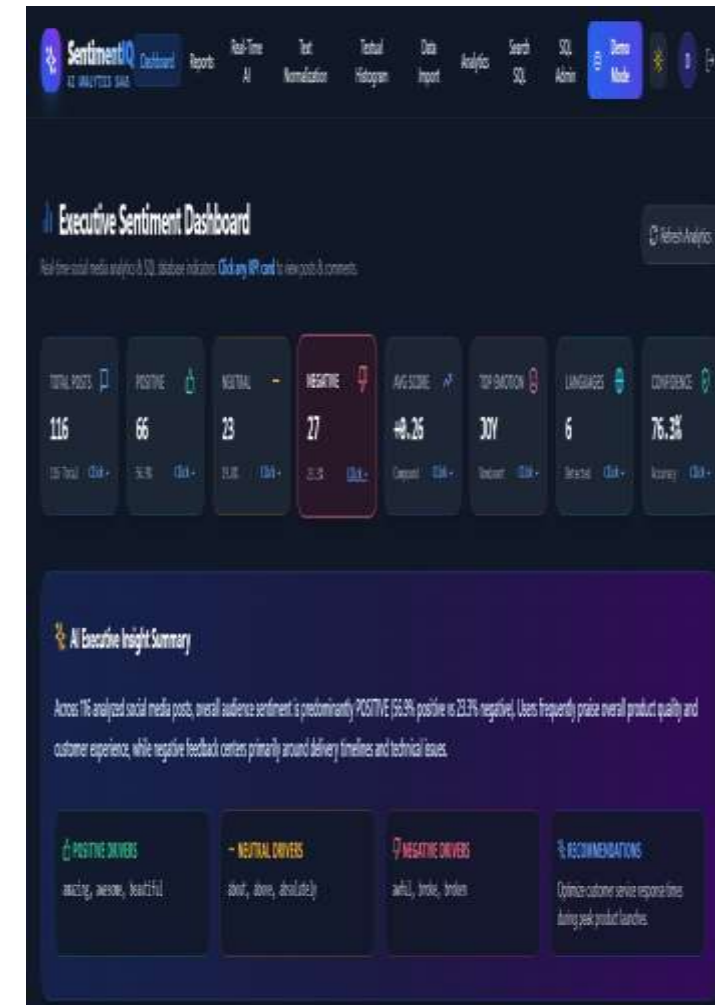


SYSTEM ARCHITECTURE



Module 1: Social Media Text Acquisition and Normalization

- Collects social media text from **command-line input, text files, or sample datasets**.
- Converts the collected text into **lowercase** to maintain uniformity during analysis.
- Removes **URLs, user mentions, hashtags, and other unwanted elements** from the text.
- Eliminates **special characters, emojis, and unnecessary symbols** to reduce text noise.
- Removes **extra spaces and formatting inconsistencies** from the collected content.
- Generates **clean and normalized text** that is ready for sentiment classification.



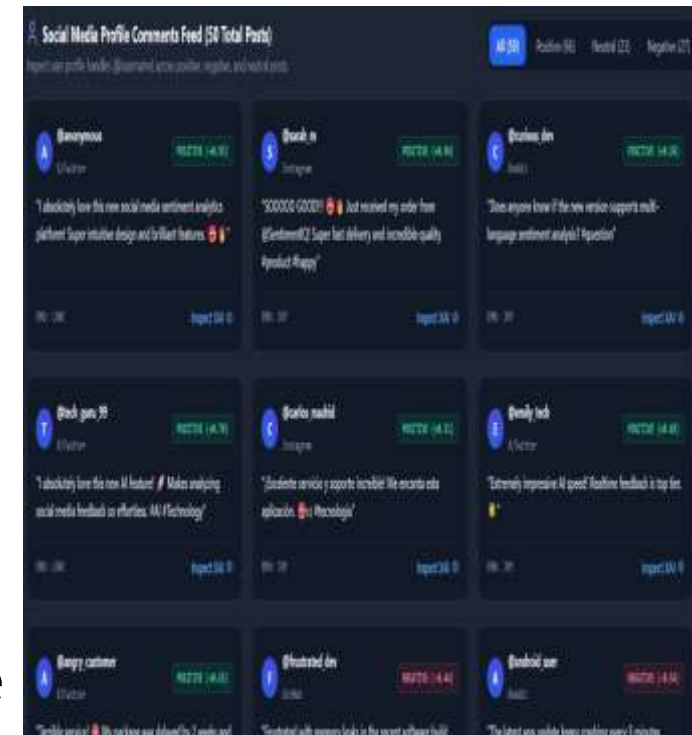
Module 2: Sentiment Analysis and Textual Histogram Generation

- Analyzes the **normalized text** to identify its overall sentiment.
- Classifies each text into **Positive, Negative, or Neutral** categories.
- Assigns a **sentiment label and score** based on the analysis.
- Counts the number of texts belonging to each **sentiment category**.
- Generates a **textual histogram** using symbols to represent sentiment distribution.
- Displays the sentiment results clearly for **easy interpretation and comparison**.



Module 3: SQL Integration and Result Management

- Stores the **sentiment analysis results** in an SQL database for organized data management.
- Creates structured tables to maintain **text, sentiment labels, and sentiment scores**.
- Uses SQL queries to **insert, retrieve, update, and manage** sentiment records.
- Enables quick retrieval of stored results for **analysis and reporting**.
- Maintains sentiment data in a **structured and reliable format**.
- Provides efficient **result management and database integration** for the complete system.
- **Generates summarized reports** from stored sentiment data for easy review and future analysis.



FUTURE SCOPE

- **Real-Time Analysis** – Analyze live social media posts for instant sentiment results.
- **Advanced AI Models** – Use machine learning and deep learning for better accuracy.
- **Multilingual Support** – Extend the system to analyze multiple languages.
- **Emotion Detection** – Detect emotions such as happiness, anger, sadness, and fear.
- **Interactive Dashboard** – Provide graphical dashboards for easy visualization of sentiment trends.
- **Large-Scale Processing** – Support large volumes of social media data efficiently.

REFERENCES

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- Liu, B. – *Sentiment Analysis and Opinion Mining*, Morgan & Claypool Publishers.
- Bird, S., Klein, E., & Loper, E. – *Natural Language Processing with Python*, O'Reilly Media.
- Pang, B., Lee, L., & Vaithyanathan, S. – *Thumbs Up? Sentiment Classification Using Machine Learning Techniques*.
- Manning, C. D., Raghavan, P., & Schütze, H. – *Introduction to Information Retrieval*, Cambridge University Press.
- Python Documentation – Text processing, command-line programming, and data handling.
- SQLite Documentation – SQL database storage, queries, and result management.

CONCLUSION

- The proposed system provides an **automated approach** for analyzing social media sentiment.
- Text normalization improves the **quality and consistency** of the input data.
- Sentiment classification identifies **positive, negative, and neutral opinions** effectively.
- The textual histogram provides a **simple and clear representation** of sentiment distribution.
- SQL integration enables **organized storage and efficient management** of analysis results.
- The command-line interface makes the system **simple, fast, and user-friendly**.
- The system can be **extended for real-time analysis and larger social media datasets** in future applications.

The background of the image is an abstract composition of flowing, wavy, and layered textures in shades of brown, gold, and dark charcoal. These textures resemble marbled paper, liquid swirls, or perhaps the grain of aged wood. The overall effect is one of organic movement and depth. Centered over this background is the text "Thank you" in a clean, white, sans-serif typeface. The text is slightly shadowed, giving it a three-dimensional appearance as if it's floating or resting on the textured surface.

Thank you